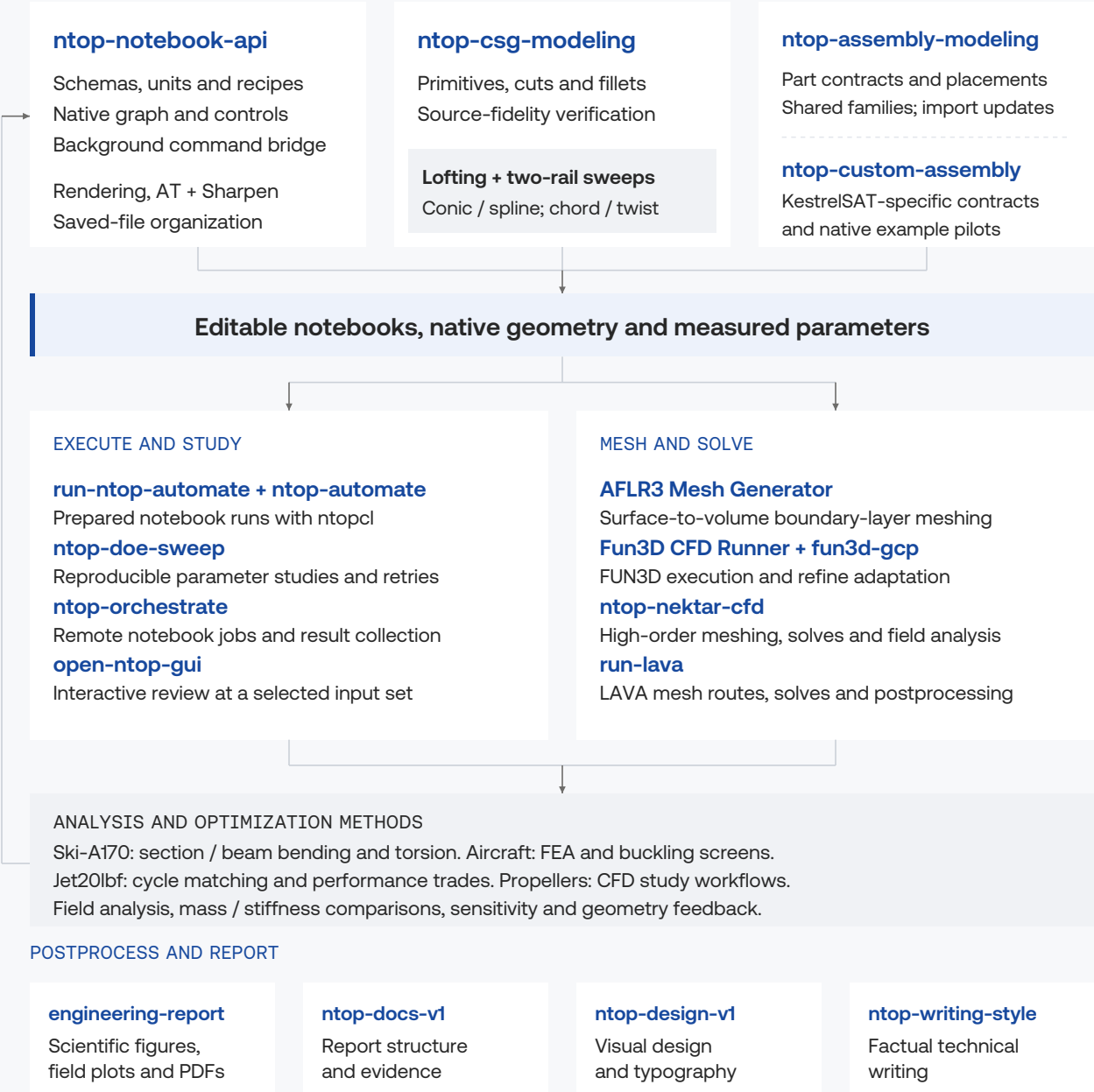


Engineering skill system

Native models, notebook execution, analysis and evidence.

18 NAMED SKILLS / INVENTORY: 16 SEP 2026

AUTHOR AND MODEL



Verification at every stage: source, graph, geometry, mesh, solver and result checks.

Blue names: skills. Grey panels: methods with case-specific evidence and limits.

Future engineering skills

Coordinated expertise around a shared aircraft design.

10 PROPOSED BRIEFS / NOT INSTALLED OR VALIDATED

Shared aircraft design contract

Mission, geometry, interfaces, materials, loads, mass, units and coordinate frames.
Every result retains assumptions, sources, revision identity and uncertainty.

George Irving configurator

Capture George-reviewed configuration rules.
Generate coordinated aircraft layouts.
Expose editable design choices and constraints.

Expert airframing

Define spars, frames, ribs and longerons.
Coordinate joints and structural load paths.
Check access and removable skin panels.

Propulsion and installation

Match engine or propulsor to the mission.
Model cycle, power and installed losses.
Coordinate inlets, exhausts and cooling.

Weights and mass properties

Maintain component mass and growth ledgers.
Calculate CG, inertia and loading cases.
Track uncertainty and configuration changes.

Stability and control (S&C)

Estimate derivatives, trim and static margins.
Assess modes and control authority.
Map limits across the flight / CG envelope.

Loads and aeroelasticity

Build maneuver, gust and ground load cases.
Transfer distributed loads into the structure.
Screen stiffness coupling and flutter risk.

Structural sizing and joints

Size members, fasteners and connections.
Check stress, buckling, fatigue and damage.
Retain allowables and failure-mode coverage.

Aerodynamics and mission

Build polars and performance models.
Compare range, endurance and field performance.
Assess model fidelity and sensitivity.

Manufacturing and assembly

Evaluate forming, machining and printing routes.
Define tolerances, tooling and assembly access.
Estimate process-driven mass, cost and risk.

Verification and trade studies

Enforce shared constraints across disciplines.
Run benchmarks and uncertainty studies.
Rank candidates with traceable evidence.

Reviewed results can update the shared design and drive new notebook variants.

Dashed boxes: proposals requiring bounded pilots, expert review and documented limits.